Solutions to Review Questions for Lecture 15

1. The box has 12 tickets, with 6 tickets labeled with -4 and the other 6 tickets labeled with the numbers $1-6$. The average of the box is $(-4*6+1+2+3+4+5+6)/12 = -0.25$ and the standard deviation $\sqrt{(16*6+1+4+9+16+25+36)/12-(-0.25)^2} = 3.94$. You are drawing 8 tickets with replacement.

2. The box has 6 tickets, with 5 tickets labeled with 0 and 1 ticket labeled with 1, representing the 5. The average of the box is $1/6 = 0.17$ and the standard deviation of the box is $\sqrt{1/6*5/6} = 0.37$. You are drawing 8 tickets with replacement.

3. The box has 2 tickets, with 1 ticket marked 0 and 1 ticket marked 1 for tails. The average of the box is $1/2 = 0.5$ and the standard deviation is $\sqrt{1/2*1/2} = 1/2$. You are drawing 100 tickets with replacement.

4. The chance of leaving the room is $1/2+1/6*1/2 = 7/12$. Hence the box would have 7 tickets labeled 0 and 5 tickets labeled 1. The average of the box is $5/12$ and the standard deviation is $\sqrt{5/12*7/12} = 0.49$. You are dring 100 tickets with replacement.

5. The box has $10,000$ tickets labeled 1 for the green M&Ms and $30,000$ tickets labeled 0 for the non-green M&Ms. The average of the box is $10,000/40,000 = 1/4$ and the standard deviation is $\sqrt{1/4*3/4} = 0.43$. You are drawing 30 tickets without replacement.

6. The box has 12 tickets, with tickets for $1, 3, 5$, representing landing heads and rolling an odd number, $0, 0, 0$ for landing heads and rolling an even number, and 6 values of -4, representing losing \$4. The average of the box is $(1+3+5+(-4)*6)/12 = -1.25$ and the standard deviation is $\sqrt{(1+9+25+6*16)/12-(1.25)^2} = 3.06$. You are drawing 16 tickets with replacement.

7. The box has 10 tickets, with 3 tickets labeled 0.01, 2 tickets labeled 0.05, 4 tickets labeled 0.10, and 1 ticket labeled 0.25. The average of the box is $(3*0.01+2*0.05+4*0.10+1*0.25)/10 = 0.078$. and the standard deviation is $\sqrt{(3*0.0001+2*0.0025+4*0.01+0.0625)/10-(0.078)^2} = 0.069$. You are drawing 4 tickets without replacement.

8. The box will have 10 tickets, with 7 tickets labeled 1 for the made free throws and 3 tickets labeled 0 for the missed free throws. The average of this is $7/10 = 0.7$ and the standard deviation $\sqrt{7/10*3/10} = 0.46$. You are drawing 2 tickets with replacement.

9. See problem 6 for the box model. The expected value is $-1.25*16 = -20$ and the standard error $\sqrt{16}*3.06 = 12.24$.

10. See problem 7 for the box model.

    a) The expected value is $4*0.078 = 0.312$.

    b) No, you can not use the formulas from class because the draws are without replacement.

    c)

```
winnings = c()
for (i in 1:10000) {
  newwinnings = sum(sample(c(rep(0.01,3),rep(0.05,2),rep(0.10,4),0.25),4,replace=FALSE))
  winnings = c(winnings,newwinnings)
  }
print(sd(winnings)*sqrt(9999/10000))
```

```
## [1] 0.1118963
```

11. The expected value is $2 * 0.7 = 1.4$ and the standard error is $\sqrt{2} * 0.46 = 0.65$.

12. a) The number of draws becomes 100.
    b) The expected value is $100 * 0.7 = 70$ and the standard error $\sqrt{100} * 0.46 = 4.6$.
    c) The expected value is $70/100 = 70\%$ and the standard error $4.6/100 = 4.6\%$.
    d) This value is `1-pbinom(64,100,0.7)=0.88`.
    e) The continuity correction means that we will use 64.5 or more instead of 65 or more because 64.5 rounds to 65. The standard units will be $z = (64.5 - 70)/4.6 = -1.20$. The normal approximation gives a probability of `1-pnorm(-1.20)=0.88`.
    f) The values are the same to 2 decimal places.

13. a) The chance of winning at least once is $1 - (5/6)^3 = 91/216$. This problem is then like 20 draws with replacement from a box with 91 tickets labeled 1 and $216 - 91 = 125$ tickets labeled 0. The expected value for the percentage will be the average of the box, or $91/216 = 0.42 = 42\%$.
    b) The draws are with replacement, and so we can use the formulas in class.
    c) The standard error is $\sqrt{(91/216 * 125/216)/20} = 0.11 = 11\%$.

14. a) The chance of winning is $3/6 = 1/2$ and so the box has 1 ticket labeled 1 and 1 ticket labeled 0. The expected value for the percentage is the average of the box, or 50%.
    b) The draws are with replacement and so we can use the formulas in class.
    c) The standard error is $\sqrt{(0.5 * 0.5)/20} = 0.11 = 11\%$.

15. a) $P(0\ green) = 0.9^4 = 0.66$.
    b) The expected value for the number of green balls is $0.1 * 4 = 0.4$ and the standard error is $\sqrt{4 * 0.1 * 0.9} = 0.6$. The standard units will be $(0.5 - 0.4)/0.6 = 0.17$ and the normal approximation gives `pnorm(0.16)=0.57`.
    c) It does not, since $4 * 0.1 = 0.4 < 5$.
    d) The exact chance will be `pbinom(35,400,0.1)-pbinom(29,400,0.1)=0.19`. The expected value for the number of green balls is $400 * 0.1 = 40$ and the standard error is $\sqrt{400 * 0.1 * 0.9} = 6$. The standard units will be $(29.5 - 40)/6 = -1.75$ and $(35.5 - 40)/6 = -0.75$ and the normal approximation gives `pnorm(-0.75)-pnorm(-1.75)=0.19` after rounding. This example follows the rule of thumb since $400 * 0.1 = 40 > 5$ and $400 * 0.9 = 360 > 5$.

16. If we let b denote the big number and s the small number and suppose that n_b tickets in the box have the big number and n_s the small number, then the average of the box is $(bn_b + sn_s)/(n_b + n_s) = bp_b + sp_s$, where p_b and p_s are the fractions with b and s. The average of the squares will similarly be $b^2p_b + s^2p_s$. Combining these gives the standard deviation as $\sqrt{b^2p_b + s^2p_s - (bp_b + sp_s)^2} = \sqrt{b^2p_b + s^2p_s - b^2p_b^2 - 2bp_bsp_s - s^2p_s^2}$. If we note that $p_b = 1 - p_s$ and $p_s = 1 - p_b$ and simplify, then the standard deviation is $\sqrt{b^2(p_b - p_b^2) + s^2(p_s - p_s^2) - 2bp_bsp_s} = \sqrt{b^2p_b(1 - p_b) + s^2p_s(1 - p_s) - 2bp_bsp_s} = \sqrt{b^2p_bp_s + s^2p_bp_s - 2bp_bsp_s}$. Further simplification gives the standard deviation as $\sqrt{p_bp_s(b^2 + s^2 - 2bs)} = \sqrt{p_bp_s(b - s)^2} = (b - s)\sqrt{p_bp_s}$.